

An over-collateralised lending protocol on Arc, with a second risk model for currency pairs.

IN ONE PARAGRAPH

Deposit an asset, borrow another against it, get liquidated if your health factor drops below 1 — the familiar shape of a lending market. What we add: risk parameters resolve **per currency pair**, not per asset. A USDC ↔ EURC loan is priced off EUR/USD volatility; anything touching WETH keeps conservative Standard parameters. One pool, one liquidation engine, two parameter sources.

Deployed on	Arc Testnet · chainId 5042002
Core contract	LendingPool.sol · 1,091 SLOC
Assets live	USDC · EURC · WETH

Why Arc specifically

USDC is native and regulated-first on Arc, with EURC alongside it as a first-class asset. That makes Arc the chain where credit *between two Circle currencies* is a natural product, not a bridged workaround.

Why it matters

A EUR/USD pair moves far less than a crypto pair, so it safely supports a much higher LTV. Existing markets don't capture that — and the mechanism generalises to further pairs.

Built for agents

Market discovery, pricing and risk checks are all read-only calls — built for autonomous strategies, not just human clicks.

Lending on Arc, extended to currency-pair credit.

USDC ↔ EURC borrowing at **90% LTV**. Liquidation parameters are computed from EUR/USD volatility by the risk engine, and validated against real positions on Arc Testnet across repeated runs.

Every risk read is a view call, so an **AI agent can drive the whole loop** without simulating the protocol.

\$105K

ON-CHAIN TVL

10

LIVE POSITIONS

175

TESTS PASSING

6

AGENT VIEW APIS

arc-lending.vercel.app · github.com/wangqiji9/arc-fx-lending · Arc Testnet chainId 5042002

A currency pair moves far less than a crypto pair.

That lower volatility is real collateral headroom — it supports a **materially higher LTV**, if the protocol actually measures it.

THE INSIGHT

EUR/USD risk is an order of magnitude smaller

ANNUALISED VOL	TYPICAL
ETH / USD	60 - 80%
BTC / USD	40 - 60%
EUR / USD	7 - 11%

Roughly **6–8× less volatile** than what a lending market normally underwrites. Charging a EUR/USD loan the same collateral cost as an ETH loan leaves most of that headroom unused.

BUT NOT ZERO

EUR/USD, measured properly

Calm regime	7–8%
Stress (2022)	~11%
Implied vol 2022	10.6%
Implied vol 2024	7.3%
Worst single day	> 2%

Lower than crypto, but a 2% gap day still happens. The higher LTV has to be **derived from these numbers**, not simply set as high as the category allows.

So we made the currency pair a first-class risk object. A registered pair gets parameters computed from its own volatility, and everything else falls back to Standard. USDC ↔ EURC is the first entry — **additional currency pairs plug into the same mechanism** as configuration, not new risk code.

One pool, two risk models.

Risk parameters resolve **per currency pair**, not per asset. Same collateral, different rules, depending on what you borrow.

STANDARD MODE

Cross-currency & volatile assets

Sourced from per-asset AssetConfig. Covers WETH collateral, shorts, and any pair without a registered FX category.

Max LTV	75%
Liquidation threshold	80%
Liquidation bonus	5.0 - 7.5%
Reserve factor	10%

FX E-MODE

USDC ↔ EURC, both directions

Sourced from FxCategory, keyed by the currency pair. Each value is computed from that pair's own measured volatility.

Max LTV	90%
Liquidation threshold	94%
Liquidation bonus	2.5%
Reserve factor	10%

Capital efficiency where it's earned

90% LTV on FX pairs vs 75% standard — **1.2× more borrowing power**, because the risk genuinely is lower.

Honest where it isn't

94%, not 98%. We give up headline LTV to keep a **liquidation window that actually works** — next slide.

WHY 94% AND NOT 98%

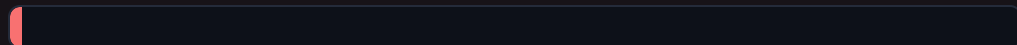
Three constraints pin the parameters.

LT and bonus are not tuned by feel. Each is bounded by a closed-form condition — and 98% violates all three.

- 1 · Death-spiral bound $\text{bonus} < (1 - \text{LT}) / \text{LT} = (1 - 0.94) / 0.94 = 6.38\% \rightarrow \text{we use } 2.5\% \checkmark$
- 2 · Healthy window $1 - \text{LT} \times (1 + \text{bonus}) = 1 - 0.94 \times 1.025 = 3.65\% \rightarrow 90\times \text{ wider } \checkmark$
- 3 · Collateral buffer $1 / \text{LT} - 1 = 1 / 0.94 - 1 = 6.38\% \rightarrow \sim 9 \text{ stress } \sigma \checkmark$

LT 98% / BONUS 2% — THE NAIVE HIGH-LTV CHOICE

HEALTHY LIQUIDATION WINDOW



0.04%

Between "liquidatable" and "already insolvent" there is **4 basis points** of price. A liquidator has no profitable window; the position goes straight to bad debt. And 2% bonus is already *at* the death-spiral bound — every liquidation pushes HF further down.

LT 94% / BONUS 2.5% — FX-DERIVED

HEALTHY LIQUIDATION WINDOW



3.65%

A **3.65% price band** in which liquidation is profitable and the protocol stays solvent. That absorbs a 2%+ FX gap move with room left over, and the 6.38% collateral buffer covers roughly nine stress-day sigmas.

The trade is deliberate: 4 points of headline LTV bought a 90× wider survival window. Full derivation in docs/architecture.md.

The close factor moves with the position.

A fixed 50% close factor leaves dust positions that no one will ever liquidate. Ours switches to full close before the position can cross into insolvency.

HEALTH FACTOR	CLOSE FACTOR	RATIONALE
$HF \geq 1.0$	NONE	Not liquidatable
$1.0 > HF \geq \text{threshold}$	50%	Partial — minimum force needed to restore health
$HF < \text{threshold}$	100%	Full close before the healthy window runs out

Standard threshold HF 0.980

FX E-Mode threshold HF 0.983

The FX threshold is higher on purpose: a narrower window means the switch to full close has to fire earlier.

Isolated by pair

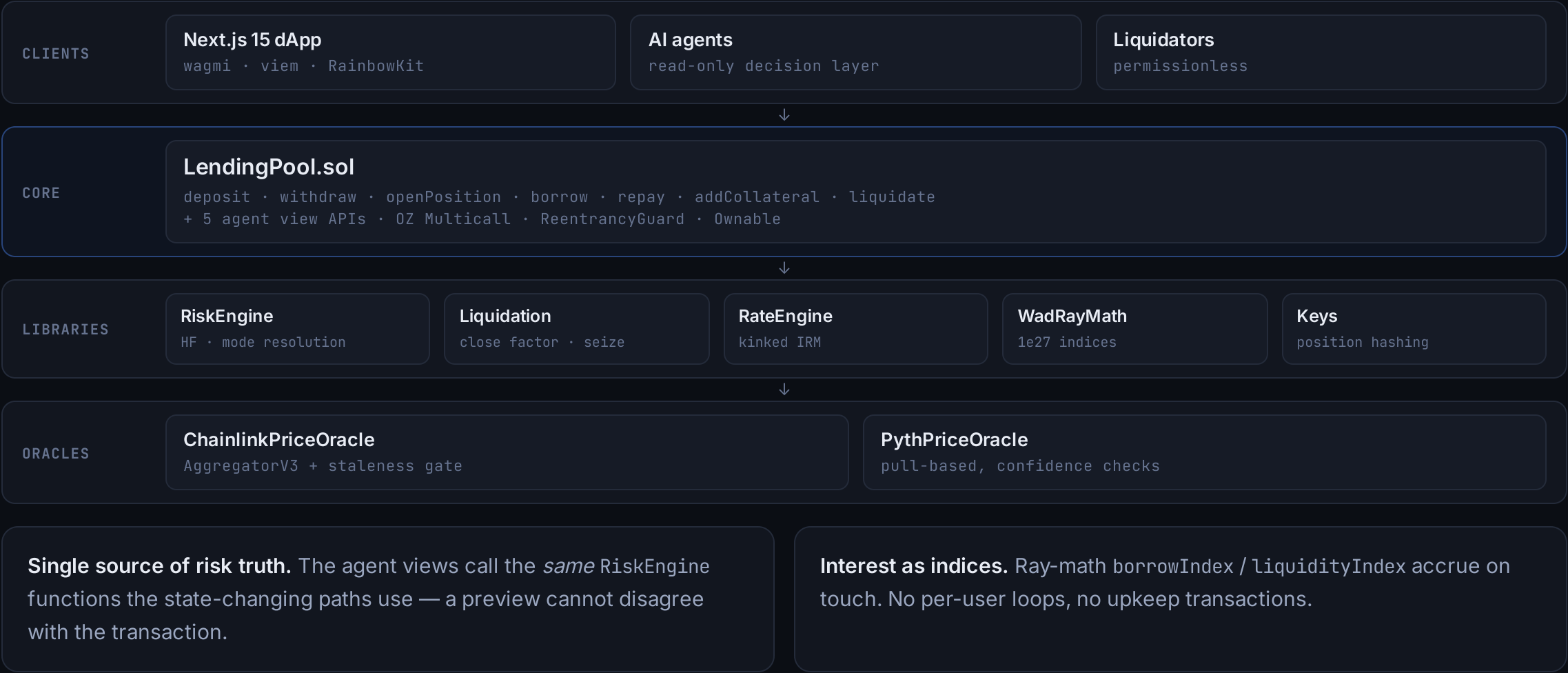
Every position is keyed by (owner, collateral, debt). One pair going bad cannot touch the borrower's other positions — contagion is bounded by construction.

Oracle pause gate

Stale price feeds block anything that *increases* risk — open, borrow, withdraw collateral, liquidate. Repay, deposit and add-collateral stay open so borrowers can always defend themselves.

One pool contract. Pure-library risk math.

1,091 lines of Solidity. No proxies, no external keepers, no off-chain risk service.



Built to be driven by an agent.

An autonomous FX strategy needs to discover markets, price a trade and check its own risk — without simulating the protocol. Five view calls cover the whole loop.

CALL	ANSWERS	RETURNS
<code>getAvailableMarkets()</code>	Which (collateral, debt) pairs can I act on, at what resolved LTV, cost and liquidity?	<code>MarketInfo[]</code>
<code>previewPosition(...)</code>	If I post X and borrow Y — will it open, at what HF, at what post-trade borrow rate?	<code>PreviewResult</code>
<code>getPositionRisk(key)</code>	How close to liquidation am I, and what is my real carry right now?	<code>PositionRisk</code>
<code>batchGetPositionRisk(...)</code>	Same, for a whole book — one RPC round trip.	<code>PositionRisk[]</code>
<code>viewRates(asset)</code>	Current borrow and supply rate, reserve factor already applied.	<code>(uint, uint)</code>
<code>multicall(bytes[])</code>	Batch <code>addCollateral</code> + borrow atomically — rebalance in one tx.	<code>bytes[]</code>

An honest signal, not a comfortable one

In Standard mode the views return a scalar `liquidationPrice`. In FX E-Mode they return a **sentinel 0 with `liquidationPriceApplicable = false`** and report `bufferBps` instead.

Because FX risk is a *depeg jump*, not a continuous drift toward a number. Handing an agent "you're still far from 1.0421" would manufacture false safety. We refuse to emit a number we don't believe.

Multicall, safely

OZ `Multicall` has a known high-severity footgun: a payable entry point lets one `msg.value` be replayed across every batched call.

The protocol has **no payable entry points** — the invariant that makes batching safe. It is documented at the contract head and guarded by a regression test, so a future Pyth fee-forwarding change can't silently reintroduce it.

Not a prototype. Real state, verifiable now.

\$./script/status.sh

chainId 5042002

POOL RESERVES	supplied	borrowed	util
USDC	61,004.27	39,104.75	64%
EURC	30,000.73	10,180.82	33%
WETH	4.0006	2.5507	63%

TVL

61,004 USDC + 32,401 (EURC@1.08) + 12,002 (WETH@3000)

= \$105,405

BORROW RATES (live, kinked IRM)

USDC	borrow 4.21%	supply 2.43%
EURC	borrow 3.70%	supply 1.13%
WETH	borrow 3.19%	supply 1.83%

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FX E-MODE

3

STD LONG

2

STD SHORT

Open positions — all three risk paths			
PAIR	MODE	DEBT	HF
WETH/USDC	STD	10,200.87	1.205
WETH/USDC	STD	6,150.65	1.202
WETH/USDC	STD	6,000.00	1.200
USDC/EURC	FX	4,180.82	1.207
USDC/EURC	FX	3,500.00	1.243
USDC/EURC	FX	2,500.00	1.253
EURC/USDC	FX	8,000.00	1.269
EURC/USDC	FX	8,251.09	1.280
USDC/WETH	STD	1.50	1.422
USDC/WETH	STD	0.90	1.481

LendingPool	0x6fc50Bbd...C9df97
PriceOracle	0x39375eD4...b7380B
USDC	0xe94C3c12...61e91f
EURC	0x657ff693...f1367C
WETH	0x78F1D761...ee02FB

PRODUCT

The whole loop works in a browser.

Connect · lend · open an FX position · watch HF · repay · liquidate — no CLI required. arc-lending.vercel.app

Borrow Markets FX E-Mode · Higher LTV

MARKET	BORROW APY	AVAILABLE	LIQ. THRESHOLD	
USDC → EURC E-Mode FX E-Mode · LTV 90%	Borrow APY 3.70%	Available 19820 EURC	Liq. Threshold 94%	Borrow
EURC → USDC E-Mode FX E-Mode · LTV 90%	Borrow APY 4.21%	Available 21900 USDC	Liq. Threshold 94%	Borrow

Standard Markets

MARKET	BORROW APY	AVAILABLE	LIQ. THRESHOLD	
USDC → WETH Standard · LTV 75%	Borrow APY 3.19%	Available 1.45 WETH	Liq. Threshold 80%	Borrow
EURC → WETH Standard · LTV 75%	Borrow APY 3.19%	Available 1.45 WETH	Liq. Threshold 80%	Borrow
WETH → USDC Standard · LTV 75%	Borrow APY 4.21%	Available 21900 USDC	Liq. Threshold 80%	Borrow
WETH → EURC Standard · LTV 75%	Borrow APY 3.70%	Available 19820 EURC	Liq. Threshold 80%	Borrow

Both risk models, one list. FX E-Mode pairs are promoted to 90% LTV / 94% LT; anything routed through a volatile leg stays at 75% / 80%. Read from the chain at render time.

Open Position FX E-Mode

Deposit collateral and borrow in one step

Collateral
10000
Bal: 19300 EURC

Borrow
5000
Max (9720 USDC) Avail: 21900 USDC

Risk Mode
FX E-Mode Standard
FX E-Mode: LTV 90%, threshold 94%, 2.5% liquidation bonus. Risk shown as safety buffer.

PREVIEW
Health Factor **+2.03**
Borrow APY **4.61%**
Safety Buffer **103.0%**
Can Open **Yes ✓**

Approve amount (EURC) **Exact**
10000 (exact)

Approve EURC

The full risk computation, itemised. HF, borrow APY and safety buffer — every figure from the same RiskEngine the transaction runs.

Positions + Supply + Borrow

Your lending deposits and borrowing positions.

Lending 2 assets

USDC Deposit Supply APY: 2.43%
Balance **1002.0027 USDC** +2.0027 USDC
Deposit Value **1002.0027 USDC** | Earned **+2.0027 USDC** | Supply APY **2.43%** | Wallet Balance **49500 USDC**
Withdraw USDC **0.00** Max
Withdraw USDC

WETH Deposit Supply APY: 1.83%
Balance **0.0501 WETH** +0.0001 WETH

Borrowing 2 positions

WETH → USDC Debt 502.1395 USDC \$502.14
Coll Debt 0.25 WETH - \$750.00 +1.19

A user's positions at a glance. Each listed separately with its own HF, collateral and accrued interest.

The risk model is the product, so it gets audited like one.

TEST SUITE

175

20

2.7×

PASSING

SUITES

TEST:SRC RATIO

✓ Invariants — solvency and index monotonicity under random op sequences

✓ Fuzz — HF and close-factor boundaries, rounding direction

✓ Scenario — full depeg / liquidation cascade walkthroughs

✓ Adversarial — oracle pause bypass, fee-on-transfer tokens, multicall value replay, reserve-factor accrual

1,091 SLOC src · 2,945 SLOC test · forge test green, 0 skipped

SELF-AUDIT LEDGER

34 findings, tracked to disposition

Five review passes over the protocol, each finding logged with an ID, a severity, a disposition (*fixed* / *wontfix* / *acknowledged*) and — where fixed — the regression test that pins it.

Math & rounding	M-1 ... M-7
Design decisions	D-1 ... D-5
Test-gap findings	T-1 ... T-21
Agent layer	A-1 ... A-3
Frontend / scripts	F, S series

Rounding always favours the protocol: debt cells, collateral floors. Every asymmetry is deliberate and test-pinned rather than incidental.

Known limits, stated plainly. Testnet oracles are mock-fed at fixed prices; production needs live Chainlink/Pyth EUR/USD with the staleness and confidence gates already implemented. Governance is a single ownable owner — a timelock is required before mainnet. No third-party audit yet.

The FX category generalises.

Everything above is one FxCategory entry. Adding a currency pair is configuration, not new risk code.

NEAR TERM

- **Live EUR/USD feeds** — swap mock oracle for Chainlink + Pyth, both adapters already written and tested
- **More stablecoin FX pairs** — each is a category registration with its own volatility-derived LT and bonus
- **Timelocked governance** — parameter changes behind a delay

BEYOND

- **Safety module** — staked backstop for the residual depeg-jump tail the parameters can't absorb
- **Agent strategies** — reference carry / rebalance bots on the existing view layer, no protocol change needed
- **Third-party audit** before any mainnet deployment

Why Arc

Arc is the chain where USDC and EURC are both first-class. FX credit between two Circle-issued currencies belongs here — and it only works if someone prices the exchange rate risk honestly instead of pretending a euro is a dollar.

LIVE APP

arc-lending.vercel.app

SOURCE

github.com/wangqiji9/arc-fx-lending

NETWORK

Arc Testnet • 5042002